Assessment Menu: Teacher Guide

The Transversal Bistro · design notes, grading and answer key

Why students cannot just pick the easy ones

1. **Points are capped by category.** Students must complete exactly 4 / 4 / 2 dishes, so they cannot fill the total with Appetizers.
2. **Chef's Specials carry 40% of the grade** and Entrées another 40%. A student who only masters the basics tops out at 20 points.
3. **Skill coverage rule.** Their 10 picks must include at least one dish from each skill group below, so every strand of the unit is assessed.
4. **Harder dishes reward process.** Each has a rubric that gives credit for correct steps, so a partly correct attempt at a harder problem earns more than a perfect easy one in the same time.

Skill group	What it covers	Dishes
PAIRS	Angle pairs formed by a transversal	A1, A2, E5, S4
ALGEBRA	Solve for angles with algebra	A6, E1, E2, S2
TRIPOLY	Triangle and polygon angle sums	A3, A4, E3, E4, S3
PROVE	Prove lines are parallel	A5, E6, S1

If a student only does…	Maximum
Only Appetizers (4 × 5)	20
Appetizers + Entrées (4 × 5 + 4 × 10)	60
Full meal (adds 2 Chef's Specials)	100

Grading and options

- Grade only the marked dishes using the rubrics on the next pages (partial credit is built in). Total possible: **100**. If a student marks more than the required number, grade the first in each category.
- Check the skill rule on the order form before totaling. If a skill group is missing, apply your own policy (for example, the dish that broke the rule scores 0, or the student may swap one dish).
- Optional "second helping": allow one extra dish per category that replaces the student's lowest score in that category.
- Look at *which* dishes students chose. Students who select only the easiest dish in each skill group may need a conference.

Student	Appetizers (of 20)	Entrées (of 40)	Specials (of 40)	Skills OK?	Total

Answer Key: Appetizers (5 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
A1 PAIRS	Parallel lines are cut by a transversal. Two angles lie **between** the parallel lines on **opposite sides** of the transversal, and one measures 72°. Name this pair and find the other angle.	**Answer:** Alternate interior angles; the other angle is 72°. Between the lines and on opposite sides: **alternate interior angles**. Parallel lines make alternate interior angles congruent, so the other angle is 72°.	2 pts: Names alternate interior angles 1 pts: States they are congruent 2 pts: Other angle 72°
A2 PAIRS	Parallel lines are cut by a transversal. Two same-side interior angles are shown; one measures 116°. Find the other angle and explain why.	**Answer:** 64°, because same-side interior angles are supplementary. Same-side interior angles add to 180° when the lines are parallel. 180 − 116 = 64°	2 pts: Uses supplementary (sum 180°) 2 pts: Correct 64° 1 pts: Gives the reason
A3 TRIPOLY	Find the sum of the interior angles of a pentagon.	**Answer:** 540° Sum = 180(n − 2) with n = 5 180(3) = 540°	2 pts: Uses 180(n − 2) 2 pts: Substitutes n = 5 1 pts: Answer 540°
A4 TRIPOLY	In a triangle, the two remote interior angles measure 35° and 48°. Find the exterior angle at the third vertex.	**Answer:** 83° Exterior angle theorem: the exterior angle equals the sum of the two remote interior angles. 35 + 48 = 83°	2 pts: States or uses the exterior angle theorem 3 pts: Correct 83°
A5 PROVE	Lines a and b are cut by a transversal, and a pair of alternate interior angles both measure 58°. Are a and b parallel? Name the theorem that tells you.	**Answer:** Yes. By the converse of the Alternate Interior Angles Theorem. The alternate interior angles are congruent (58° = 58°). By the **converse of the Alternate Interior Angles Theorem**, the lines are parallel.	1 pts: Says yes 2 pts: Notes the angles are congruent 2 pts: Names the converse of the alternate interior angles theorem
A6 ALGEBRA	Parallel lines are cut by a transversal. Two corresponding angles measure (3x + 10)° and 70°. Find x.	**Answer:** x = 20 Corresponding angles are congruent: 3x + 10 = 70 3x = 60, so x = 20	2 pts: Sets the angles equal 1 pts: Names corresponding angles are congruent 2 pts: x = 20

Answer Key: Entrées (10 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
E1 ALGEBRA	Parallel lines are cut by a transversal. Same-side interior angles measure (4x + 8)° and (2x + 10)°. Find x and both angle measures.	**Answer:** x = 27; the angles are 116° and 64°. Same-side interior angles are supplementary: (4x + 8) + (2x + 10) = 180 6x + 18 = 180, so x = 27 4(27) + 8 = 116° and 2(27) + 10 = 64° (sum 180 ✓)	3 pts: Sets the sum equal to 180 2 pts: Combines terms; x = 27 3 pts: Both angle measures 2 pts: Reason: same-side interior angles are supplementary
E2 ALGEBRA	Parallel lines are cut by a transversal. Alternate interior angles measure (5x − 12)° and (3x + 14)°. Find x and the measure of each angle.	**Answer:** x = 13; each angle is 53°. Alternate interior angles are congruent: 5x − 12 = 3x + 14 2x = 26, so x = 13 5(13) − 12 = 53° and 3(13) + 14 = 53°	3 pts: Sets the angles equal 2 pts: x = 13 3 pts: Both angles 53° 2 pts: Reason named
E3 TRIPOLY	An exterior angle of a triangle measures (7x − 5)°. The remote interior angles measure (2x + 15)° and (3x + 10)°. Find x and the exterior angle.	**Answer:** x = 15; the exterior angle is 100°. Exterior angle theorem: 7x − 5 = (2x + 15) + (3x + 10) 7x − 5 = 5x + 25, so 2x = 30 and x = 15 7(15) − 5 = 100°; check 45 + 55 = 100 ✓	3 pts: Sets exterior = sum of remote interior 2 pts: Solves x = 15 3 pts: Exterior angle 100° 2 pts: Reason named
E4 TRIPOLY	Each interior angle of a regular polygon measures 156°. How many sides does the polygon have? Show your method.	**Answer:** 15 sides Each exterior angle is 180 − 156 = 24°. Exterior angles of any polygon add to 360°, so n = $\frac{360}{24}$ = 15.	3 pts: Exterior angle 24° 3 pts: Uses the 360° exterior sum 2 pts: n = 15 2 pts: Explains the method
E5 PAIRS	Lines m and n are parallel and cut by a transversal. ∠2 and ∠7 are alternate exterior angles and m∠2 = 125°. Lee says m∠7 = 55° “because angles along a transversal add to 180°.” Find his mistake and give the correct m∠7.	**Answer:** Alternate exterior angles are congruent, not supplementary. m∠7 = 125°. Lee treated the pair as supplementary, but that is only true for same-side pairs. ∠2 and ∠7 are **alternate exterior angles**, which are congruent when lines are parallel. m∠7 = 125°	4 pts: Identifies Lee used the wrong relationship 3 pts: States alternate exterior angles are congruent 3 pts: Correct m∠7 = 125°
E6 PROVE	Lines a and b are cut by a transversal. Alternate interior angles measure (3x + 20)° and (5x − 40)°. Find x so that a and b are parallel, and justify your answer with a theorem.	**Answer:** x = 30 (both angles are 110°). For parallel lines the alternate interior angles are congruent. 3x + 20 = 5x − 40, so 60 = 2x and x = 30 3(30) + 20 = 110° and 5(30) − 40 = 110° Congruent alternate interior angles make the lines parallel by the converse of the Alternate Interior Angles Theorem.	3 pts: Sets the angles equal 2 pts: x = 30 2 pts: Both angles 110° 3 pts: Cites the converse theorem

Answer Key: Chef’s Specials (20 pts each)

Worked solutions and rubrics. Graphs: draw on the number line; endpoints as noted.

Dish	Problem	Solution	Rubric
S1 PROVE	Lines a and b are cut by transversal t. ∠1 and ∠3 are alternate exterior angles and ∠1 ≅ ∠3. ∠2 is vertical to ∠1, and ∠2 and ∠3 are corresponding angles. Write a proof, with a reason for each step, that a and b are parallel.	**Answer:** a and b are parallel (converse of the Corresponding Angles Postulate). 1. ∠1 ≅ ∠3 (given) 2. ∠1 ≅ ∠2 (vertical angles are congruent) 3. ∠2 ≅ ∠3 (transitive property of congruence) 4. ∠2 and ∠3 are congruent corresponding angles, so a and b are parallel (converse of the Corresponding Angles Postulate)	4 pts: ∠1 ≅ ∠2 with reason: vertical angles 5 pts: ∠2 ≅ ∠3 with reason: transitive or substitution 6 pts: Concludes a and b are parallel by the converse of the corresponding angles postulate 2 pts: States the given 3 pts: Logical order, every step has a reason
S2 ALGEBRA	Lines m and n are parallel and cut by transversal t. Alternate interior angles ∠1 and ∠2 measure (3x + 7)° and (5x − 25)°. **(a)** Find x and m∠1. **(b)** ∠3 forms a linear pair with ∠2, and ∠4 corresponds to ∠3 with m∠4 = (5y − 25)°. Find m∠3, y, and m∠4.	**Answer:** (a) x = 16, m∠1 = 55° (b) m∠3 = 125°, y = 30, m∠4 = 125° (a) 3x + 7 = 5x − 25, so 2x = 32 and **x = 16**; m∠1 = 3(16) + 7 = **55°** (m∠2 = 55° too) (b) Linear pair: m∠3 = 180 − 55 = **125°** Corresponding angles are congruent: 5y − 25 = 125, so 5y = 150 and **y = 30** m∠4 = 5(30) − 25 = **125°**	3 pts: (a) Sets alternate interior angles equal 3 pts: (a) x = 16 2 pts: (a) m∠1 = 55° 3 pts: (b) m∠3 = 125° by linear pair 3 pts: (b) Sets 5y − 25 = 125 3 pts: (b) y = 30 3 pts: (b) m∠4 = 125°
S3 TRIPOLY	In triangle ABC, m∠A = (2x + 10)°, m∠B = (3x − 5)°, m∠C = (x + 25)°. **(a)** Find x and all three angles. **(b)** Extend CB past B to point D. Find the exterior angle ∠ABD in **two different ways** and show they agree.	**Answer:** (a) x = 25; ∠A = 60°, ∠B = 70°, ∠C = 50° (b) ∠ABD = 110° both ways (a) Sum is 180: 6x + 30 = 180, so **x = 25**; A = 60°, B = 70°, C = 50° (b) Linear pair: ∠ABD = 180 − 70 = **110°** Exterior angle theorem: ∠ABD = m∠A + m∠C = 60 + 50 = **110°** Both methods give 110°.	4 pts: (a) Sets the sum to 180 3 pts: (a) x = 25 3 pts: (a) Three angles correct 4 pts: (b) Linear pair method 110° 4 pts: (b) Exterior angle theorem method 110° 2 pts: (b) States they agree
S4 PAIRS	Parallel lines m and n are cut by a transversal. At m the angles are ∠1 (upper left), ∠2 (upper right), ∠3 (lower left), ∠4 (lower right); at n they are ∠5, ∠6, ∠7, ∠8 in the same positions. m∠1 = 130°. **(a)** Find all eight angles. **(b)** Name the pair type for ∠1 and ∠8, ∠4 and ∠5, and ∠3 and ∠5.	**Answer:** (a) ∠1, ∠4, ∠5, ∠8 = 130°; ∠2, ∠3, ∠6, ∠7 = 50° (b) alternate exterior; alternate interior; same-side interior (a) ∠4 is vertical to ∠1 and ∠2, ∠3 form linear pairs: ∠4 = 130°, ∠2 = ∠3 = 50° Corresponding angles are congruent: ∠5 = 130°, ∠6 = 50°, ∠7 = 50°, ∠8 = 130° (b) ∠1 and ∠8: **alternate exterior** (130° and 130°) ∠4 and ∠5: **alternate interior** (130° and 130°) ∠3 and ∠5: **same-side interior** (50° + 130° = 180°)	2 pts: (a) ∠4 = 130° 3 pts: (a) ∠2 = ∠3 = 50° 5 pts: (a) ∠5 to ∠8 correct 3 pts: (b) Alternate exterior 3 pts: (b) Alternate interior 3 pts: (b) Same-side interior 1 pts: (b) Notes equal or supplementary